Understanding the Phase in Single-Polarization SAR Images: A Ship Classification Study

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***Abstract*—In this article, an in-depth analysis is presented on the role of the phase in the ship classification using single-polarization synthetic aperture radar (SAR) images. To conduct the analysis, a novel lightweight convolutional neural network (CNN) model, LeNet-2AP, is introduced. Based on the LeNet-5 architecture, the proposed model processes a two-channel amplitude and phase input through two distinct and independent branches, which are merged only at the final stage before the softmax layer. Several modern enhancements, including weight standardization and global average pooling (GAP), are incorporated to improve the performance and efficiency. The model is evaluated on the OpenSARShip dataset using single-look complex (SLC) samples and tested against amplitude–phase, in-phase and quadrature-phase (I–Q), and complex-valued inputs, achieving the superior performance compared with existing models in the literature. Consequently, Shapley additive explanations (SHAP) is applied to analyze the model's decisions, focusing on the contribution of the phase to the classification process. It is demonstrated that the phase information is distributed across an extensive area beyond the ship's amplitude signature in the SAR image. Furthermore, the phase is found to provide a comparable level of discriminative information to the amplitude, challenging the traditional approach of using the amplitude exclusively. These findings highlight the significance of the phase as a complementary source of information in single-polarization SAR images and demonstrate that shallow, optimized CNNs are better suited to efficiently extract this information with limited data samples.**

***Index Terms*—Deep learning, phase, ship classification, synthetic aperture radar (SAR).**

I. Introduction

IN AN increasingly globalized world, where roughly 80% of global trade relies on shipping, maritime domain awareness (MDA) has become a critical area of focus [1]. Traditional vessel monitoring tools, while valuable, face significant limitations. For instance, the automatic identification system (AIS) depends on the voluntary cooperation of vessels [2]. Therefore, ships involved in illicit activities, such as illegal fishing or evading international sanctions, can easily operate undetected by disabling their AIS transponder [3]. Similarly, maritime patrols are both costly and limited in their range, particularly beyond coastal zones [4]. In such scenarios, the utility of space-based synthetic aperture radar (SAR) becomes evident. SAR does not only provide extensive coverage but also enables persistent monitoring through cloud cover and during nighttime, capabilities that optical sensors lack [5]. The ability to classify ships in SAR images holds substantial value, offering a detailed understanding of maritime traffic that is consistent with MDA requirements. When multiple potential targets are detected, information on ships' classes allow for a more timely and prioritized response [6].

For surveillance tasks, single-polarization SAR is typically employed [7]. Although dual- and quad-polarimetric SAR can provide additional information, they require more advanced systems that are not as readily available [8]. In single-polarization acquisitions, amplitude data are almost exclusively utilized. Contrastingly, the phase, often deemed random and uninformative, is generally discarded. This perception stems from the fully developed speckle model [9], which describes the scattering behavior of radar waves from rough surfaces relative to the radar wavelength. It is based on the following assumptions.

1) *Large Scatterer Population:* Each resolution cell contains numerous independent scatterers, with no single scatterer dominating.
2) *Statistical Independence:* The magnitude and phase of each scatterer are statistically independent, both from each other and from other scatterers.
3) *Uniform Phase Distribution:* The phases of the scatterers are uniformly distributed over the interval $(-\pi, \pi]$.

These assumptions were applicable for low-resolution SAR sensors, where objects were typically represented as point targets. However, the advent of high-resolution systems has revealed their limitations when imaging extended man-made structures. As demonstrated in [10], such targets exhibit a phenomenon known as sideband response. Its origin can be understood by examining the demodulated baseband signal of a point target S_{bb}, which is expressed as follows:

$$S_{\text{bb}}(\tau, \eta) = A \exp(j\psi) R(\tau, \eta) \tag{1}$$

where A and ψ are the amplitude and phase of the backscattered signal, respectively, j is the imaginary unit, R is the impulse response of the SAR system, τ is the fast time, and η is the slow time.

In the conventional SAR theory, ψ is assumed constant during the radar exposure time t_e for a given target [11]

$$\frac{d\psi}{dt} = 0, \quad \text{for } 0 \le t \le t_e. \tag{2}$$

However, man-made targets predominantly contain intricate structures with cavity-like reflectors that trap the incident wave

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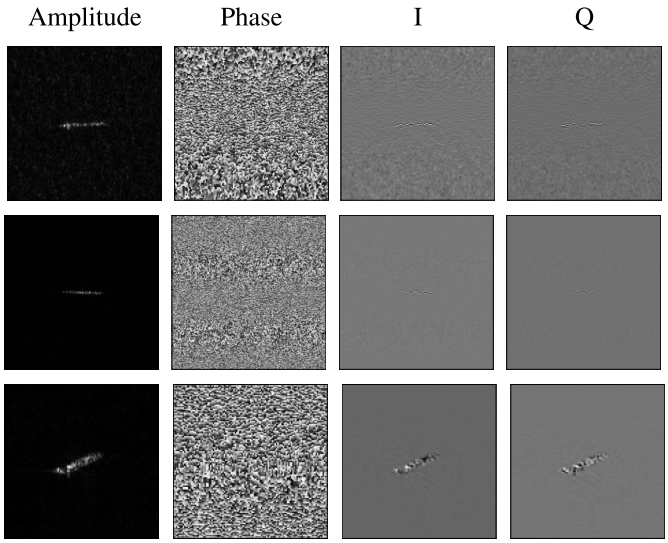


Fig. 1. Ship samples from the OpenSARShip dataset were displayed in amplitude, phase, I, and Q formats. The first, second, and third rows correspond to a bulk carrier, a container ship, and a tanker, respectively.

before reflecting it back. This interaction introduces a time delay that modulates both amplitude and phase. While the amplitude is typically linearly modulated, the phase modulation can be either linear or nonlinear [12]. The former causes effects such as translation or smearing in the SAR image. In contrast, nonlinear modulation results in a nonlinear delay at the phase center. This leads to unintended artifacts in the SAR processor's output, which are the aforementioned sideband responses. The single-look complex (SLC) SAR image retains these artifacts through its preservation of the phase, whereas it is lost in the ground range detected (GRD) image [13].

Sideband responses exhibit several distinct characteristics that differentiate them from range and Doppler sidelobes. They are among the strongest responses, are not confined to specific range or cross-range gates, and extend over a region significantly larger than the actual target area [10]. Notably, since these responses originate from the target itself, they cannot be mitigated using conventional weighting techniques. These unique properties have motivated research into the use of the phase for both detection and classification tasks in single-polarization SLC images.

A. Related Works

Existing approaches utilizing the phase can be broadly divided into two categories: statistical-based methods and deep learning-based methods. Statistical methods involve modeling the phase or in-phase and quadrature-phase (I–Q) distributions of SAR images to manually extract phase-related features. On the other hand, deep learning approaches primarily use convolutional neural networks (CNNs), either with a two-channel input of amplitude–phase or I–Q images (see Fig. 1) or with a single-channel complex-valued input. Given the focus of this article on single-polarization images, research exploiting the phase for dual- and quad-polarimetric SAR has been excluded for conciseness.

1) Statistical Methods: El-Darymli et al. [13] proposed an algorithm using the Hilbert spectral analysis to detect dispersive nonlinear scatterers in MSTAR target chips. The method also characterized the order of nonlinearity. In [9], an in-depth investigation of nonlinear dynamics in complex-valued SAR images was presented. It demonstrated that nonlinearity is statistically significant for extended targets but not for point targets. It was also shown that power-detected images retain some nonlinear artifacts, albeit in a diminished form, unlike magnitude-detected images, where these artifacts are entirely removed. Furthermore, nonlinearity was found to depend on the ratio of the target's dimensions to the image's spatial resolution, with larger ratios producing more pronounced effects.

In [14], complex-valued statistics were applied using circularity/noncircularity analysis. Target-free SAR images were shown to be second-order circular, rendering their phase uninformative. Contrastingly, target-containing images exhibited high noncircularity, which was found to be proportional to nonlinearity. In addition, El-Darymli et al. [12] presented the backscatter relative phase image (BRPI), where the phase of a pixel was related to its neighbors through convolutional kernels. This framework statistically modeled the phase using the wrapped complex Gaussian scale mixture. Moreover, a set of 15 phase-based features were tested, alongside power-based features, on eight ground target classes from the MSTAR dataset using a support vector machine (SVM). The results showed significant classification improvements through complementary information provided by the two sets of features.

Leng et al. [15] applied normalized noncircularity and non-Gaussianity to the maritime domain, discriminating ships from radio frequency interferences (RFIs). Due to distinct scattering mechanisms, ships exhibited high noncircularity and sub-Gaussian I–Q distributions. On the contrary, RFIs and sea clutter displayed low noncircularity and Gaussian or sub-Gaussian distributions, respectively. In [16], the complex signal kurtosis (CSK) was introduced as a metric accounting for both noncircularity and non-Gaussianity. An algorithm based on the CSK outperformed the constant false alarm rate (CFAR) method for ship detection on Sentinel-1, TerraSAR-X, and Gaofen-3 images, leveraging the phase to minimize false alarms. The CSK was also used to measure ship signature quality, improve defocused signatures, and suppress sidelobes [17]. Adding to that, Hinai and Guida [18] explored the relationship between the CSK and ship characteristics such as length, orientation angle relative to the sensor, and look angle. Contributions of ship sections, such as the deckhouse, to the CSK were examined using high-resolution Capella Space images.

Zhao and Liu [19] proposed a method using noncircularity and similarity variance-weighted information entropy (SVWIE). The method enhanced ship targets and suppressed sea clutter, which was modeled using a generalized gamma distribution, in ALOS-2 SLC images. Wu et al. [20] derived three parameters from the complex generalized Gaussian distribution to classify natural and man-made urban targets in submeter resolution airborne SAR data. Similarly, Wu et al. [21] introduced three variables exploiting noncircularity for the same classification task. These included differences in the mean, skewness, and kurtosis of the I–Q distributions.

2) Deep Learning Methods: Gleich and Sipos [22] employed a complex-valued CNN (CVCNN) to classify

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TerraSAR-X spotlight images into three land cover classes: urban, vegetation, and water. The model was composed of two convolutional, pooling, and fully connected layers and was tested using varying input sizes. In [23], a CVCNN with complex-valued batch normalization and squeeze-and-excitation module was proposed. It was used to classify ten target classes from the MSTAR dataset and outperformed real-valued SENets by 2%. Zeng et al. [24] proposed a three-channel CVCNN, where images were processed using three distinct convolutional kernels to capture multiscale features. The channels were fused using three methods: averaging, concatenation, and taking the maximum. Tested on MSTAR data, the network outperformed real-valued models, with ablation studies highlighting the importance of the multistream, multiscale approach and fusion strategies.

Wang et al. [25] proposed data augmentation techniques tailored for complex-valued images, including a complex-valued mean filter, complex Gaussian noise, and subaperture decomposition. A CVCNN incorporating complex-valued residual modules was shown to outperform ResNet18 on the MSTAR dataset. The overall classification accuracy improvement was around 1.5%, at the expense of a fourfold increase to the training time. The improvement increased to 5% when training on half the available data.

Deng et al. [26] presented a sparse representation of SLC images, enhancing targets while retaining phase information. A real-valued two-channel CNN, taking as input amplitude–phase images, was tested on the enhanced MSTAR images. It was shown that the incorporation of the phase improved the overall classification accuracy by less than 0.5%. Song et al. [27] also presented a novel sparse recovery algorithm and proposed the use of a real-valued three-channel CNN to classify MSTAR targets. The three channels received as an input the amplitude, I, and Q images. When compared with a single channel amplitude input, the average performance increase was around a percentage point.

Zhang et al. [28] introduced CV-MotionNet for classifying three classes of moving ships in simulated and Gaofen-3 images. Similarly, CV-RotNet was presented for classifying defocused ship targets [29]. It achieved more than a 2% increase in the overall classification accuracy compared with its real-valued counterpart when tested on a three-class Gaofen-3 dataset. Zhao et al. [30] proposed a CVCNN based on VGG16 called CVGGNet, which was tested on bulk carriers, container ships, and tankers from the OpenSARShip dataset. A main contribution was the introduction of an area-based max pooling as a better alternative to amplitude-based pooling. The model achieved a 1% and 0.3% increase in the overall classification accuracy compared with real-valued VGG16 and ResNet18 models, respectively. The study also investigated multiple complex-valued activation functions, such as the hyperbolic tangent (tanh), the exponential linear unit (ELU), and the rectified linear unit (ReLU), demonstrating that the latter performed best.

B. Contributions

The recent literature has demonstrated that there is additional value in incorporating the phase into CNNs for SAR classification tasks. However, two main challenges arise. First, many of the presented models are based on CVCNNs. While their ability to enhance the classification accuracy has been demonstrated, they result in a substantial increase in computation. Although a few studies have proposed the use of real-valued multiple channel CNNs as an alternative, the lack of SAR-specific adaptations in these approaches limits their ability to effectively capture the unique characteristics of the phase. Second, the vast majority of studies focus solely on classification performance. Despite its importance, this approach overlooks the critical need for interpretability. Including the phase exacerbates the already existing challenges of understanding CNN decision-making. Without a thorough analysis, knowledge regarding its role in the classification process and its interaction with the amplitude remains incomplete.

In response to these critical issues, this article introduces LeNet-2AP, a lightweight CNN adapted from LeNet-5, to effectively capture the unique characteristics of phase data in the SAR ship classification. To address interpretability, an in-depth analysis is conducted using the proposed network in conjunction with Shapley additive explanations (SHAP) [31]. This approach elucidates the contribution of the phase to the classification process in relation to the amplitude, offering valuable insights into the network's decision-making.

The main contributions of this article are as follows.

1) Demonstrating that the phase in single-channel SAR images is useful for ship classification when appropriately integrated.
2) Proposing a novel lightweight CNN architecture, LeNet-2AP, designed to maximize the complementary use of amplitude and phase.
3) Presenting a comprehensive explainability analysis using SHAP to characterize the behavior of phase contributions across ship classes through aggregate-level metrics.
4) Introducing an efficient SHAP method using the LeNet-2AP architecture to quantify amplitude and phase contributions via global average pooling (GAP)-level analysis.

The remainder of this article is structured as follows. Section II highlights the methodology, describing the model architecture and interpretability analysis. Section III details the experimental setup, including the utilized dataset and the classification process. Section IV presents the classification results, ablation study of LeNet-2AP, and interpretability analysis using SHAP, along with a discussion of these findings. Section V provides concluding remarks, highlighting the limitations of the study and future extensions of the research.

II. Methodology

A. Proposed Model

1) Model Formulation: It has been presented earlier that the two main approaches to incorporating the phase into CNNs are through complex-valued operations or by using multiple real-valued channels. While complex-valued operations naturally handle both amplitude and phase within a single representation, they significantly increase the number of computations required. This can be shown by contrasting standard

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and complex-valued operations in CNNs. The standard 2-D convolutional operation is defined as follows:

$$(\mathbf{H} * \mathbf{K})\left(l_f, m_f\right) = \sum_{l_k} \sum_{m_k} \mathbf{H}\left(l_f - l_k, m_f - m_k\right) \mathbf{K}\left(l_k, m_k\right) \quad (3)$$

where $\mathbf{H}$ is the input, $\mathbf{K}$ is the convolutional kernel, (l_f, m_f) are the spatial coordinates of the output feature map, and (l_k, m_k) are the spatial coordinates of the kernel.

For a complex input $\mathbf{H}_z = \mathbf{H}_r + \mathbf{H}_j j$ and complex kernel $\mathbf{K}_z = \mathbf{K}_r + \mathbf{K}_j j$, the convolution becomes

$$(\mathbf{H}_z * \mathbf{K}_z) = \left(\mathbf{H}_r * \mathbf{K}_r - \mathbf{H}_j * \mathbf{K}_j\right) + \left(\mathbf{H}_r * \mathbf{K}_j + \mathbf{H}_j * \mathbf{K}_r\right) j. \quad (4)$$

Similarly, a real-valued dense layer performs the operation

$$\mathbf{u} = \mathbf{W}\mathbf{x} + \mathbf{b} \quad (5)$$

where $\mathbf{u}$ is the output vector, $\mathbf{W}$ is the weight matrix, $\mathbf{x}$ is the input vector, and $\mathbf{b}$ is the bias vector.

For a complex input $\mathbf{x}_z = \mathbf{x}_r + \mathbf{x}_j j$, weight $\mathbf{W}_z = \mathbf{W}_r + \mathbf{W}_j j$, and bias $\mathbf{b}_z = \mathbf{b}_r + \mathbf{b}_j j$, the complex dense layer operation becomes

$$\mathbf{U}_z = \left(\mathbf{W}_r \mathbf{x}_r - \mathbf{W}_j \mathbf{x}_j + \mathbf{b}_r\right) + \left(\mathbf{W}_r \mathbf{x}_j + \mathbf{W}_j \mathbf{x}_r + \mathbf{b}_j\right) j. \quad (6)$$

Hence, any convolutional and dense operations in the complex domain require double the computations when compared with two real-valued channels. In addition, while deep learning libraries like PyTorch and TensorFlow have started introducing complex-valued operations, their support remains limited to basic functions, such as convolutions, while lacking more advanced features like normalization techniques and dense layers. These limitations necessitate custom implementations, which often lack the optimization and reliability of established frameworks.

For two real-valued channels, either I–Q or amplitude–phase images can be used. While theoretically equivalent in the information they contain, the two formats differ in their impact on interpretability. The I–Q format implicitly encodes both amplitude and phase and is the sensor's inherent output. However, the amplitude–phase format aligns more directly with the physical properties captured by SAR. This explicit representation also aids in distinguishing the individual contributions of amplitude and phase to the classification process, making it especially useful in applications where interpretability is essential.

Building on the topic of interpretability, the choice of deep CNN architectures presents notable disadvantages. Such networks are typically required to capture the complexity of large datasets with many classes, as it is common in object classification tasks like those in ImageNet [32]. However, given the current task and the limited amount of available data, a shallow CNN architecture is arguably more suitable. This approach not only greatly improves the traceability of feature extraction but also significantly mitigates the risk of overfitting [33]. While advanced methods for transfer learning and fine-tuning deep CNNs exist to adapt them to limited data [34], [35], [36], these strategies do not resolve the interpretability challenge and instead increase the resources required for implementation.

TABLE I
LAYER CONFIGURATION AND PARAMETERS OF THE PROPOSED LENET-2AP ARCHITECTURE

Layer	Kernel Size	Output Shape	Number of Params
Input	-	(128, 128, 2)	0
Amp Slicing	-	(128, 128, 1)	0
Phase Slicing	-	(128, 128, 1)	0
Amp Conv1	5 × 5	(124, 124, 6)	156
Phase Conv1	5 × 5	(124, 124, 6)	156
Amp Max Pool	4 × 4	(31, 31, 6)	0
Phase Avg Pool	4 × 4	(31, 31, 6)	0
Amp Dropout1	-	(31, 31, 6)	0
Phase Dropout1	-	(31, 31, 6)	0
Amp Conv2	5 × 5	(27, 27, 16)	2416
Phase Conv2	5 × 5	(27, 27, 16)	2416
Amp Dropout2	-	(27, 27, 16)	0
Phase Dropout2	-	(27, 27, 16)	0
Amp Global Avg Pool	-	(16)	0
Phase Global Avg Pool	-	(16)	0
Concatenate	-	(32)	0
Dense Output	-	(3)	96

2) Model Architecture: Based on all the aforementioned considerations, a shallow, real-valued two-channel CNN, named LeNet-2AP, is proposed. Its architecture, illustrated in Fig. 2, builds on LeNet-5, which gained recognition as one of the earliest efficient implementations of CNNs [37]. Recent studies have demonstrated its utility for SAR ship classification, showing that it performs comparably to more complex networks, such as VGG19, given limited data [38]. Consequently, LeNet-5 was selected as a suitable foundation, with several enhancements introduced to improve its efficiency and adapt it to the specific characteristics of phase data.

One of the main modifications in LeNet-2AP is the implementation of a late fusion strategy for its two channels. Unlike color channels in RGB images, which are typically combined in the first convolutional layer, amplitude and phase data represent distinct sources of information. This is comparable to merging two imaging modalities, such as optical and SAR data, where it was shown that using independent branches improved classification results [39]. Consequently, LeNet-2AP processes amplitude and phase in independent branches, fusing them only before the softmax layer. This theoretically allows each branch to adapt and specialize more effectively to the distinct inputs.

The separate streams also enable customisations specific to each type of data. Modern CNNs generally favor max pooling over average pooling [40], which is advantageous for the amplitude branch as it propagates the strongest features that often hold discriminative information. However, the phase, being cyclical and constrained within a fixed range, benefits more from considering the surrounding context; hence, average pooling was chosen. To improve the computational efficiency, the pooling window size was doubled to 4×4.

In both branches, the tanh activation function was replaced with ReLU, which is widely recognized for improving training efficiency and performance [41]. Weight standardization was also applied, normalizing the weights of each convolutional kernel to have zero mean and unit variance before each

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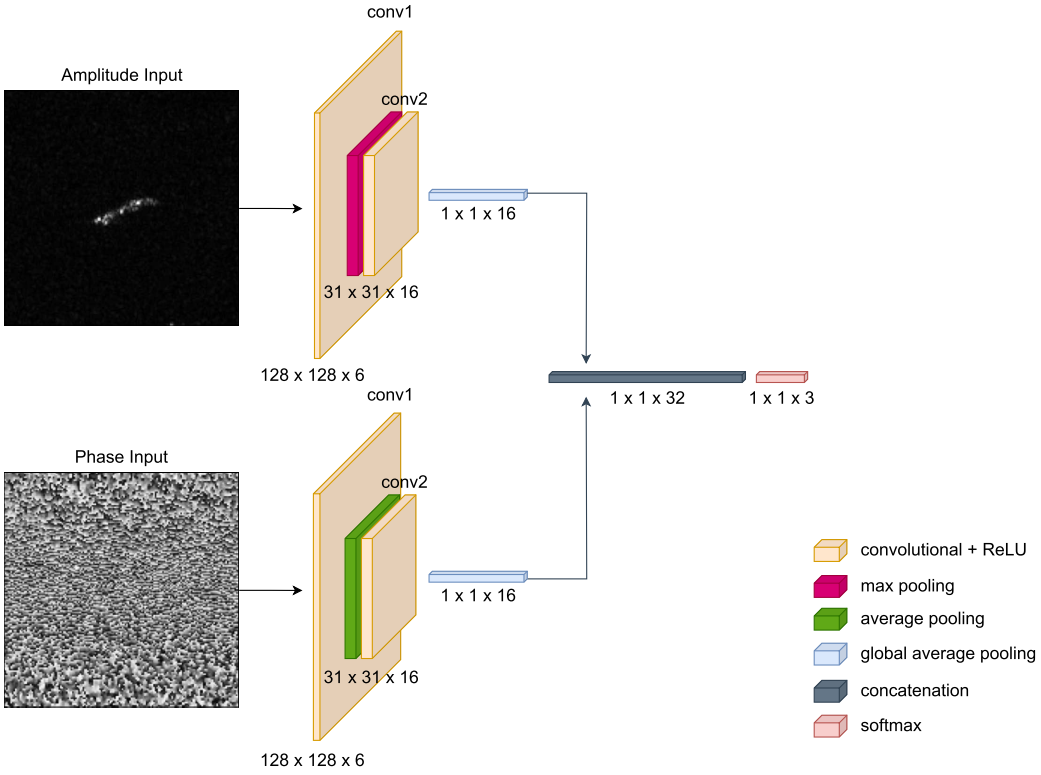


Fig. 2. LeNet-2AP architecture: a modified version of LeNet-5 used in this study.

forward pass. This technique generally improves training stability and helps the network learn more effectively by reducing the sensitivity of the model to initialization [42]. Finally, fully connected layers were replaced by GAP. Studies have shown that CNNs derive much of their representational power from the convolutional layers, making fully connected layers unnecessary for many tasks [43]. GAP provides a structural advantage by reducing the number of parameters, which not only leads to a more lightweight and efficient model but also decreases the risk of overfitting. Table I details all the layers of LeNet-2AP, together with their shape and parameter count.

B. Interpretability Analysis

1) SHAP Framework: The second part of this study focuses on an interpretability analysis to clarify the contributions of amplitude and phase. Several methodologies were suitable for this purpose, with the most widely used being gradient-weighted class activation mapping (Grad-CAM) [44], local interpretable model-agnostic explanations (LIMEs) [45], and SHAP. While all three enable model-agnostic analysis of input contributions to classification outcomes, SHAP was chosen for this study due to two main reasons. First, the literature suggests that SHAP provides more consistent and reliable interpretations for SAR man-made targets, such as vehicles [46]. Second, SHAP offers a finer level of detail by treating each pixel as an independent feature, unlike Grad-CAM, which produces broader heatmaps or LIME, which relies on super-pixels that group connected pixels with similar characteristics.

Shapley values, originating from the cooperative game theory, provide a fair method for attributing each feature's contribution to a model's output [47]. The Shapley value ϕ_i for feature i is calculated as follows:

$$\phi_i = \sum_{F_s \subseteq F \setminus \{i\}} \frac{(\#F_s)!\,(\#F - \#F_s - 1)!}{(\#F)!} \left[f(F_s \cup i) - f(F_s) \right] \quad (7)$$

where F is the full set of features, F_s ranges over all subsets of F that excludes the ith feature ($F_s \subseteq F \setminus i$), $\#F_s$ denotes the cardinality of F_s, $f(F_s)$ is the model output using F_s, $\cup$ is the union operator, and ! is the factorial.

The preceding equation calculates the marginal contribution of a feature to the model's output by evaluating all possible subsets of features. However, the number of subsets grows exponentially with the number of features, making this method computationally infeasible for images given the large pixel count. SHAP introduces a more efficient approach without performing the summation over F_s. In the context of deep learning models, SHAP values can be efficiently approximated using the Deep SHAP method, which combines SHAP values with the DeepLIFT algorithm [48]. The SHAP value $\varphi_i(f(x))$ for the ith feature of sample x with respect to the model's output $f(x)$ can approximated as follows:

$$\varphi_i(f(x)) \approx m_i \cdot (x_i - E[x_i]) \quad (8)$$

where m_i is the multiplier for feature i, computed using DeepLIFT's chain rule, x_i is the value of feature i in the input

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sample x, and $E[x_i]$ is the expected value of feature i over a set of background samples X', defined as follows:

$$E[x_i] = \frac{1}{\#X'} \sum_{x' \in X'} x'_i \tag{9}$$

where x' denotes a background sample from X'.

The set of background samples, also known as the baseline, is chosen to represent the distribution of training data. This approximation allows for efficient computation of SHAP values in deep neural networks by leveraging the model's internal structure to propagate feature contributions directly to the output. Deep SHAP recursively passes multipliers backward through the network, combining contributions from all paths involving feature i into a single SHAP value.

A key aspect of SHAP is that the sum of SHAP values across all features satisfies the additivity property

$$\sum_{i=1}^{n_f} \varphi_i = f(x) - E\left[f(x)\right] \tag{10}$$

where n_f is the total number of features in sample x.

This property ensures that SHAP values fully distribute the difference between the model's output and the baseline among the individual feature contributions, enabling the precise attribution of feature importance. For an input, SHAP generates a set of values for each class included in the prediction. Positive values indicate feature contributions supporting a given decision, while negative values represent contributions opposing it.

2) Pixel-Level Analysis: A key objective was to investigate whether the phase is spatially linked to the amplitude signature of the ship. To address this, SHAP values were used to measure the contribution magnitude of each pixel to the prediction. These values were aggregated within the minimum enclosing rectangle (MER) to compute the MER contribution ratio (MCR), which quantifies the relative importance attributed by the model to the MER compared with the background. For a specific true class y and channel c, the MCR is defined as follows:

$$\mathrm{MCR}_{y,c} = \frac{1}{n_y} \sum_{i=1}^{n_y} \left(\frac{1}{n_p} \sum_{p=1}^{n_p} \frac{\sum_{(l,m) \in S_{\mathrm{mer}}} \left|\varphi_{i,p,c}(l,m)\right|}{\sum_{(l,m)} \left|\varphi_{i,p,c}(l,m)\right|} \right) \tag{11}$$

where S_{mer} is the set of pixel coordinates within the MER, n_p is the number of predicted classes, n_y is the number of samples belonging to the same true class, and $|\varphi_{i,p,c}(l,m)|$ is the absolute SHAP value for the (l,m)th pixel in the ith sample's pth predicted class and cth channel.

To examine how the influence of amplitude and phase spreads beyond the MER, the distance contribution function (DCF) is defined. It quantifies the mean absolute SHAP value within concentric squares at distance d from the ship's center, expanding outwards by one pixel in all direction at each step. For each true class y and channel c, the DCF is computed as follows:

$$\mathrm{DCF}_{y,c}(d) = \frac{1}{n_y} \sum_{i=1}^{n_y} \left(\frac{1}{n_p} \sum_{p=1}^{n_p} \frac{1}{\#S_d} \sum_{(l,m) \in S_d} \left|\varphi_{i,p,c}(l,m)\right| \right) \tag{12}$$

where S_d is the set of pixel coordinates within the concentric square at distance d from the ship's center.

Consistent with the MCR computation, the DCF aggregates SHAP contributions across predicted classes and true-class samples, capturing how the model's attributed importance varies with increasing distance from the ship's center.

To summarize feature contributions for a specific class visually, a composite image (CI) was constructed by averaging the absolute SHAP values at each pixel (l,m) across all n_y samples belonging to that class. This CI highlights the regions with the highest contributions for each channel, offering an intuitive visual representation of the areas most influential in the model's decision-making for that class. It is defined as follows:

$$\mathrm{CI}_{y,c}(l,m) = \frac{1}{n_y} \sum_{i=1}^{n_y} \left(\frac{1}{n_p} \sum_{p=1}^{n_p} \left|\varphi_{i,p,c}(l,m)\right| \right). \tag{13}$$

Finally, to quantify the contribution of the amplitude and phase channels as a whole, the relative contribution (RC) for channel c is computed as follows:

$$\mathrm{RC}_{y,c} = \frac{1}{n_y} \sum_{i=1}^{n_y} \left(\frac{1}{n_p} \sum_{p=1}^{n_p} \frac{\sum_{(l,m)} \left|\varphi_{i,p,c}(l,m)\right|}{\sum_{c'} \sum_{(l,m)} \left|\varphi_{i,p,c'}(l,m)\right|} \right) \tag{14}$$

where c' is the indexes all channels.

3) GAP-Level Analysis: Computing pixel-level contributions using SHAP requires backpropagating the output gradients through all layers of the network. This approach requires substantial computational resources, even for a relatively shallow model like LeNet-2AP. While such computations are necessary for analyzing spatial patterns, they are less critical for evaluating the overall contributions of individual channels. To address this, a more computationally efficient method is proposed, leveraging the late fusion mechanism in LeNet-2AP.

As the model maintains separation of the amplitude and phase channels up until the softmax layer, the contribution of each branch can be evaluated using the high-level features within the GAP vector. To facilitate such an evaluation, the model is conceptually divided into two parts. The first acts as a feature extractor, transforming the image input into a 1-D vector at the GAP layer. This vector contains 32 features: the first 16 derived from the amplitude and the remaining 16 from the phase. The second part serves as the classifier, taking the GAP vector as input to produce the final classification result. While this adjustment does not affect the model's classification outcomes, it applies SHAP exclusively to the GAP and softmax layers, significantly reducing computational requirements.

Using the proposed alternative, metrics analogous to the CI and RC can be calculated within the context of the GAP vector. While CI quantifies the average contribution of each pixel per true class, the composite GAP vector (CG) instead summarizes the average contribution of each high-level feature in the GAP vector for a specific class. It is defined as follows:

$$\mathrm{CG}_y(k) = \frac{1}{n_y} \sum_{i=1}^{n_y} \left(\frac{1}{n_p} \sum_{p=1}^{n_p} \left|\varphi_{i,p}(k)\right| \right) \tag{15}$$

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where $|\varphi_{i,p}(k)|$ is the absolute SHAP value for the kth GAP element of the ith sample's pth predicted class.

Similarly, the contribution of channel c as a whole can be derived using the RC from the GAP vector (RCG), defined as follows:

$$\mathrm{RCG}_{y,c} = \frac{1}{n_y}\sum_{i=1}^{n_y}\left(\frac{1}{n_p}\sum_{p=1}^{n_p}\frac{\sum_{k\in G_c}\left|\varphi_{i,p}(k)\right|}{\sum_{k}\left|\varphi_{i,p}(k)\right|}\right) \tag{16}$$

where G_c represents the indices of GAP vector elements derived from channel c, defined as follows:

$$G_c = \begin{cases}\{1,\ldots,16\}, & \text{if } c = \text{amplitude}\\ \{17,\ldots,32\}, & \text{if } c = \text{phase.}\end{cases} \tag{17}$$

III. Experimental Setup

A. Dataset

The availability of SAR ship datasets has expanded rapidly with the rise of deep learning. However, the majority of these datasets are primarily designed for ship detection [49]. For fine-grained ship classification, which requires precise labeling of ship types, only three major datasets are widely adopted: OpenSARShip [50], OpenSARShip 2.0 [51], and FUSAR-Ship [52]. The first two are composed of Sentinel-1 images, while the latter uses Gaofen-3. However, FUSAR-Ship provides amplitude data exclusively, limiting its applicability for studies that incorporate the phase.

OpenSARShip is considered the first publicly available dataset specifically dedicated to SAR ship classification. It includes 11 346 ships captured in the interferometric wide (IW) swath mode. Samples in the dataset are provided in dual polarizations (VV and VH) and are divided between GRD and SLC formats. The SLC samples have a spatial resolution of 3×22 m in range and azimuth, respectively. OpenSARShip 2.0, released shortly after, builds on its predecessor by expanding the dataset to 34 528 samples. Nonetheless, many of the additional images feature multiple ships, strong interferences, or land structures such as ports, presenting a more complex classification scenario [53]. In order to ensure a clear representation of any phase-related results and attribute them solely to the target, the original OpenSARShip dataset is selected.

In terms of ship classes, three were considered as: bulk carriers, container ships, and oil tankers. These ship types represent a large portion of the global fleet and play a vital role in international trade [54]. Their prominence is reflected in the large number of samples they contribute to the dataset, while other classes contain an insufficient number to support reliable conclusions. In total, 333 bulk carriers, 577 container ships, and 514 tankers are used in this study. Given that previous research has shown that co-polarized data provides more distinctive information for maritime targets [55], the VV polarization has been chosen.

B. Computational Setup

All code implementations were carried out in Python, primarily using TensorFlow, Scikit-learn, and OpenCV. Model training and testing were conducted on a high-performance computing (HPC) cluster, with an allocation of 22 GB of RAM and 22 GB of GPU memory.

TABLE II
Training–Testing Split of the OpenSARShip SLC Samples

OpenSARShip SLC		
	Train	Test
Bulk carrier	233	100
Container ship	233	344
Tanker	233	281

C. Classification Process

To mitigate the class imbalance present in OpenSARShip, a stratified undersampling approach was employed to create a balanced training and validation set [56]. Specifically, 70% of the smallest class (bulk carrier) was used as a baseline, resulting in 233 samples per class for training. The remaining samples were reserved as a holdout testing set to assess the model generalization. Table II provides a breakdown of the sample distribution across classes.

The proposed model was compared with a baseline LeNet-5 for validation. The activation function in the baseline was also changed to ReLU, as the model suffered from overfitting using tanh. In addition, several models from the literature were included for evaluation, namely, CVGGNet, VGG19, ResNet50V2, Xception, DenseNet121, MobileNetV2, and NASNetMobile. This selection covers a range of architectural depths and complexities, allowing for a comprehensive comparison. Benchmark models were evaluated across four data formats: a single amplitude channel, a two-channel amplitude–phase format, a two-channel I–Q format, and a single complex-valued channel. The single-channel complex-valued format specifically refers to a CVCNN that processes complex numbers directly, using complex-valued operations such as those in (4) and (6), rather than separating them into real and imaginary components and using real-valued operations as done in the I–Q format. Due to computational limitations, only LeNet-5, VGG19, and CVGGNet were tested with the complex-valued format. LeNet-2AP was evaluated exclusively on the amplitude–phase input, as it was specifically designed for it.

Ablation studies were conducted to demonstrate the performance contribution of each added component in LeNet-2AP, such as the late fusion mechanism and customized pooling strategies. Furthermore, to isolate the classification performance of the amplitude and phase channels, each branch was tested individually by configuring the GAP vector to include features derived solely from one source.

With the exception of LeNet-2AP, LeNet-5, CVGGNet, and the complex-valued VGG19, all models were initialized with ImageNet weights and fine-tuned on OpenSARShip samples, as training from scratch resulted in overfitting. For these models, the input was rescaled to a range of -1 to 1 to align with the ImageNet format, while for the others, the input was normalized from 0 to 1. The learning rate and dropout rate were initially set to 1e$-$2 and 0.2, respectively. However, VGG19 and CVGGNet required a lower learning rate of $1e^{-6}$, as they failed to converge with the default setting. Training was conducted for a maximum of 400 epochs, using an early

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stopping callback with a patience of 20 epochs to mitigate overfitting. During fine-tuning, models were trained for up to 100 epochs with a reduced learning rate of $1e^{-6}$ and an early stopping patience of 10 epochs.

A fivefold cross-validation approach was employed, where the training data were split into five equal parts. For each fold, one part served as the validation set, while the remaining four were used for training. To account for variability from nondeterministic processes, each fold was repeated three times, and the results were averaged to ensure robustness.

D. SCR Analysis

To evaluate the model performance under varying signal conditions, an analysis based on the signal-to-clutter ratio (SCR) was conducted. This analysis required a curated subset of the testing data to ensure the quality of the SCR measurements. First, the MER of each ship in the full testing set was determined using the adaptive edge detection with a Sobel filter [57]. While this approach was effective, samples with strong interference occasionally yielded inaccurate MERs. To ensure reliability, the entire testing set was visually inspected to identify and exclude such cases. This resulted in a final analyzed testing set of 65 bulk carriers, 256 container ships, and 136 tankers. The SCR was calculated as follows:

$$\mathrm{SCR} = 10 \log 10 \left(\frac{I_{\mathrm{ship}}}{I_{\mathrm{clutter}}} \right) \tag{18}$$

where I_{ship} is the average pixel intensity within the ship's MER. The local clutter intensity, I_{clutter}, was calculated from the pixels within a surrounding border region, which was defined by a fixed margin of 10 pixels from the MER border.

E. SHAP Process

The DeepExplainer algorithm from the Python SHAP library was used for the interpretability study. The SHAP baseline comprised the complete training dataset from one of the fivefold cross-validation splits, comprising 559 samples balanced across the three ship classes. The SHAP analysis was then performed on the curated testing set of 457 samples described in Section III-D, as accurate MERs were required. This selection also had the added benefit of reducing computational load, while still providing sufficient samples for reliable and representative interpretability results.

IV. Results and Discussion

A. Performance Study

Table III presents the classification results and performance metrics of LeNet-2AP alongside various benchmark CNNs. For the amplitude input, DenseNet121 emerges as the top-performing model, achieving the highest values across all four classification metrics. Despite being relatively small in model size and number of parameters compared with VGG19, ResNet50V2, and Xception, it has the second longest training and inference times at 2.0 s and 2.5 ms, respectively. MobileNetV2 appears to offer the best balance between performance and efficiency. It achieves the second highest performance metrics that are within a 2% margin to DenseNet121, while requiring half the training and inference times and three times less the number for parameters. These results indicate that while deeper models like DenseNet121 benefit from their representational capacity, lightweight models such as MobileNetV2 can achieve competitive results when computational efficiency is a priority. While LeNet-5 displays the lowest accuracy, its training and inference times are ten times faster than the next fastest model.

For the amplitude–phase input, it can be seen that all benchmark models significantly decrease in the classification performance compared with the amplitude input, except for the accuracy of LeNet-5. An increase in accuracy but decrease in the other three classification metrics stems from the imbalanced testing set. Given that the precision, recall, and $F1$ score are calculated individually for each class and then averaged with equal weighting, the behavior of LeNet-5 indicates that the addition of the phase channel improves the classification in one of the testing classes with a high sample number, while reducing it in a minority class. This pattern is evident in the container ship and bulk carrier classes. Specifically, the $F1$ score for the container ship class increases from 64% (amplitude input) to 72% (amplitude–phase input), supported by an increase in true positives from 190 to 251, respectively. In contrast, the bulk carrier class experiences a sharp drop in performance, with the $F1$ score decreasing from 41% (amplitude input) to 19% (amplitude–phase input) and true positives declining from 56 to 23, respectively. This outcome highlights that the apparent gains from using the amplitude–phase input may result from benchmark models capturing spurious patterns in the phase data, which do not generalize consistently across classes at the testing time.

In terms of the other benchmark models, the drop in performance can be attributed to the limitations of transfer learning in this context. The amplitude channel benefits from pretrained ImageNet weights due to some shared characteristics with optical images, such as edges, textures, and contrast, allowing some degree of generalization. However, this does not extend to the phase channel, as its underlying information is fundamentally different from the visual patterns encountered in traditional object classification tasks. Instead of leveraging this additional input, the models likely treat it as noise, disrupting the learned feature extraction process and leading to the observed drop in performance. This result reinforces the limitations of transfer learning-based approaches when applied to multimodal SAR data, particularly when integrating modalities with fundamentally different information structures like phase. The smaller classification decrease in VGG19, approximately 3% compared with 8%–10% in the other models, could be attributed to the fully connected layers' ability to mitigate the impact of the phase channel to a greater extent than GAP.

The results from the I–Q input exhibit a similar trend, further supporting the previous argument. Compared with the amplitude input, all models decrease in classification performance. However, the drop is less pronounced than with the amplitude–phase input. This can be attributed to the fact that, unlike the amplitude–phase input, where the phase channel is introduced as a distinct source, the I–Q representation combines the amplitude and phase information into a unified input.

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TABLE III

CLASSIFICATION AND MODEL METRICS OF LENET-2AP AND BENCHAMRK MODELS, WHERE AMPLITUDE REPRESENTS A ONE-CHANNEL INPUT OF AMPLITUDE VALUES, AMPLITUDE–PHASE REPRESENTS A TWO-CHANNEL INPUT OF AMPLITUDE AND PHASE VALUES, I–Q REPRESENTS A TWO-CHANNEL INPUT OF I AND Q IMAGES, AND COMPLEX-VALUED REPRESENTS A ONE-CHANNEL COMPLEX-VALUED INPUT WITH COMPLEX-VALUED OPERATIONS

Model	Accuracy	Precision	Recall	F1 Score	Training Time per epoch (s)	Model Size (MB)	Number of Parameters (M)	Inference Time per sample (ms)
Amplitude								
LeNet-5	59.2%	57.2%	59.5%	56.0%	**0.1**	6.2	1.6	**0.1**
VGG19	60.9%	58.3%	58.1%	55.7%	1.3	268.5	70.4	1.0
ResNet50V2	65.0%	63.2%	65.0%	61.6%	1.2	89.9	23.6	1.5
Xception	65.0%	61.5%	64.0%	61.2%	1.4	79.4	20.8	1.4
DenseNet121	67.7%	65.5%	68.2%	64.3%	2.0	26.9	7.0	2.5
MobileNetV2	66.7%	64.8%	66.2%	62.9%	1.0	8.6	2.3	1.1
NASNetMobile	64.9%	61.5%	64.2%	61.1%	3.8	16.3	4.3	4.7
Amplitude-Phase								
LeNet-5	62.5%	54.5%	55.1%	52.3%	0.2	6.2	1.6	0.2
VGG19	58.2%	54.6%	53.8%	52.9%	1.2	268.5	70.4	0.9
ResNet50V2	57.8%	55.1%	53.9%	52.6%	1.2	89.9	23.6	1.5
Xception	55.0%	52.1%	51.3%	49.9%	1.4	79.4	20.8	1.4
DenseNet121	57.2%	55.4%	54.6%	53.0%	2.0	26.9	7.0	2.3
MobileNetV2	59.1%	57.3%	57.3%	55.2%	0.9	8.6	2.3	1.0
NASNetMobile	55.8%	53.9%	52.4%	51.4%	3.8	16.3	4.3	4.6
LeNet-2AP	**71.9%**	**67.7%**	**69.3%**	**66.9%**	0.2	**0.02**	**0.005**	0.3
I-Q								
LeNet-5	58.8%	57.0%	59.8%	55.9%	0.2	6.2	1.6	0.2
VGG19	59.0%	59.5%	58.4%	55.4%	1.2	268.5	70.4	0.9
ResNet50V2	62.4%	59.8%	61.8%	58.8%	1.2	89.9	23.6	1.9
Xception	57.8%	56.4%	57.6%	54.5%	1.3	79.4	20.8	1.4
DenseNet121	59.0%	58.1%	59.5%	55.9%	2.1	26.9	7.0	2.5
MobileNetV2	65.7%	63.8%	66.1%	62.2%	0.9	8.6	2.3	1.1
NASNetMobile	61.2%	58.9%	59.6%	57.1%	4.3	16.3	4.3	5.7
Complex-valued								
LeNet-5	61.5%	58.7%	60.0%	57.4%	0.5	12.4	3.3	0.7
VGG19	64.7%	64.7%	64.5%	61.1%	9.8	536.9	140.8	10.3
CVGGNet [30]	64.4%	63.9%	63.9%	60.6%	8.3	496.4	130.1	8.4

Consequently, the I–Q images are likely more interpretable to models pretrained on ImageNet, as the merged representation may partially align with the visual patterns and features these models are optimized to extract. Thus, it can be deduced that the implicit integration of the phase via the I–Q input offers a more compatible format for pretrained architectures, though it is still incapable of fully exploiting phase-specific features.

Using the complex-valued input, it can be seen that both LeNet-5 and VGG19 perform better than with the amplitude input. LeNet-5 increases by 2.3% in accuracy and 1.4% in $F1$ score. However, this improvement comes at a substantial cost to efficiency, as the model size and parameter count is doubled. In addition, the training and inference times increase fivefold and sevenfold, respectively, for LeNet-5. The efficiency impact is further pronounced for VGG19. With nearly 141 million parameters, training and inference times increase tenfold. Nonetheless, the improvement margin is higher, as it achieves a 3.8% gain in accuracy, and 5.4% in $F1$ score. While VGG19 slightly outperforms CVGGNet in classification metrics, CVGGNet displays better efficiency. These findings suggest that although CVCNNs can deliver modest performance gains, their significant computational demands may restrict their suitability for resource-constrained or real-time SAR applications.

Finally and most importantly, LeNet-2AP outperforms all models across various input types. Compared with the second-best model, DenseNet121 with the amplitude input, LeNet-2AP surpasses it by 4.2%, 2.2%, 1.1%, and 2.6% in accuracy, precision, recall, and $F1$ score, respectively. This performance is even more notable when considering its efficiency. Its training and inference times are an order of magnitude less than DenseNet121 and over three times faster than MobileNetV2. Furthermore, its architecture is exceptionally lightweight, with a model size of just 20 kB and approximately 5000 parameters, making it two orders of magnitude smaller in size than the real-valued LeNet-5 and three orders of magnitude lower in parameter count. This demonstrates that SAR-specific lightweight designs like LeNet-2AP are not only capable of effectively integrating phase information but can also outperform much deeper models, both in efficiency and performance. Hence, targeted architectural adaptation in SAR classification tasks is crucial to maximize the added value that the phase can provide.

B. SCR Study

Fig. 3 presents a comparison of the classification accuracy of LeNet-2AP using both amplitude and phase branches, versus the amplitude branch exclusively, across binned SCR values. The average SCR of the 457 curated testing samples was

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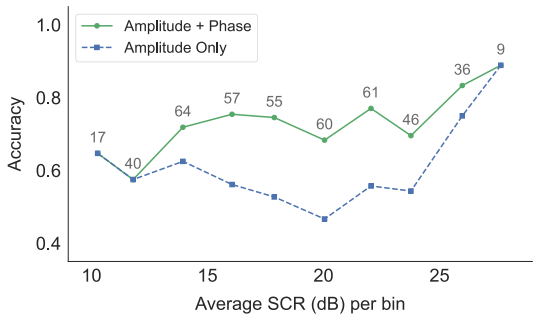


Fig. 3. LeNet-2AP accuracy as a function of the average SCR per bin. The plot compares the performance of the full model using both amplitude and phase (green solid line) against the amplitude branch exclusively (blue dashed line). Each data point represents the average accuracy within a 2-dB-wide SCR bin. The number of testing samples within each bin is annotated in gray.

calculated to be 17.8 dB. At the extreme ends of the SCR range, both models achieve similar accuracy. From 24 to 28 dB, the accuracy improves for both models, reflecting the expected benefit of a clearer target signature relative to the surrounding clutter.

However, the most notable and diverging behavior occurs in the mid-SCR range, approximately between 14 and 22 dB. In this region, the performance gap between the two models becomes pronounced. For example, in the 20-dB bin, the dual-branch LeNet-2AP achieves an accuracy of approximately 68%, while the amplitude-only model falls to around 47%. Interestingly, while the accuracy of the dual-branch model remains relatively stable across this mid-SCR range, the performance of the amplitude-only model decreases with increasing SCR. This behavior suggests that the relationship between SCR and classification accuracy is more complex than a simple monotonic improvement. Although a higher SCR typically indicates a target that is more distinguishable from the background, this does not necessarily translate to improved classification accuracy. A target may be easily detectable yet still visually or statistically ambiguous between classes, particularly if its amplitude characteristics resemble those of another category.

To quantitatively characterize this relationship, a linear fit across the SCR bins was computed. The full model exhibited an approximate 1.2% accuracy increase per decibels increase in SCR, compared with 0.9% for the amplitude-only model, suggesting that classification performance improves more rapidly with signal quality when phase information is available. The Pearson correlation coefficient between SCR and classification accuracy was calculated as 0.79, with a p-value of 0.007, indicating a statistically significant positive relationship. In contrast, the amplitude-only model yielded a weaker and nonsignificant correlation, measured at 0.43, with a p-value of 0.21, implying that amplitude features alone are less reliably influenced by improvements in SCR. Finally, the standard deviation of accuracy across SCR bins was measured at 8.5% for the full model and 11.7% for the amplitude-only variant, underscoring the phase-enhanced model's greater consistency under varying signal conditions.

Crucially, these results demonstrate that incorporating the phase not only improves overall classification accuracy but also stabilizes the model's performance across varying SCR conditions. The dual-branch LeNet-2AP shows increased robustness to SCR variability, maintaining higher and more consistent accuracy where the amplitude-only model degrades.

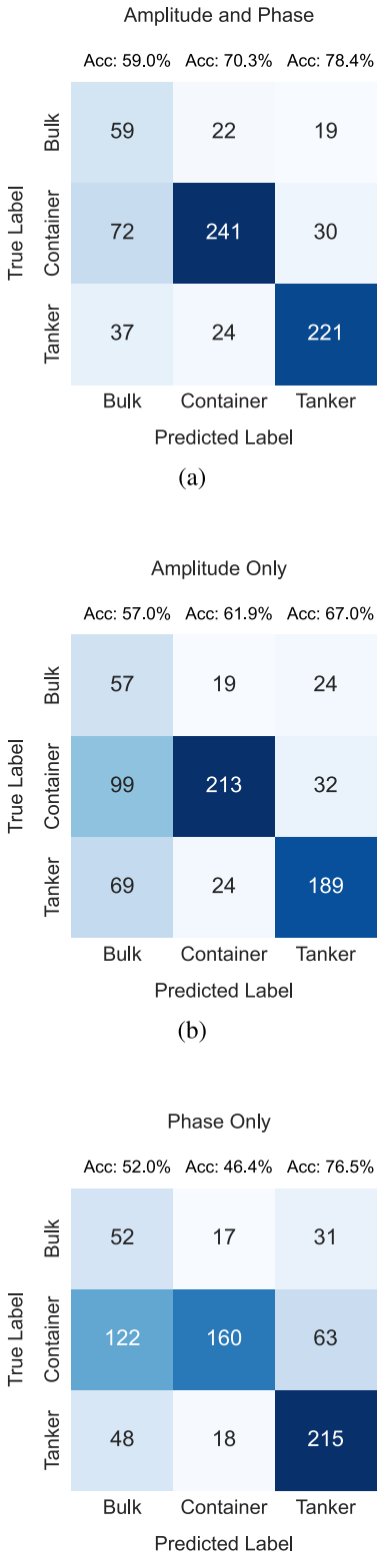


Fig. 4. Confusion matrix of predictions generated by LeNet-2AP, using (a) both amplitude and phase channels, (b) amplitude exclusively, and (c) phase exclusively.

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TABLE IV

ABLATION STUDY OF LENET-2AP, SHOWING THE IMPACT OF INDIVIDUAL COMPONENTS ON CLASSIFICATION AND MODEL METRICS. CLASSIFICATION RESULTS OF THE ABLATED COMPONENTS ARE PRESENTED AS RELATIVE CHANGES (Δ), REFERENCED TO LENET-2AP'S RESULTS

Ablated Component	Δ Accuracy	Δ Precision	Δ Recall	Δ F1 Score	Best Epoch	Model Size (MB)	Number of Parameters	Inference Time per sample (ms)
None (LeNet-2AP)	**71.9%**	**67.7%**	**69.3%**	**66.9%**	**39**	**0.02**	**5243**	**0.25**
ReLu → Tanh	-4.0%	-5.2%	-7.0%	-5.6%	87	0.02	5243	0.29
Weight Standardisation (removed)	-7.2%	-6.3%	-7.4%	-7.2%	105	0.02	5243	0.28
Global Average Pooling → Fully Connected Layers	-11.6%	-9.3%	-9.9%	-10.6%	47	10.74	2815043	0.27
Second Pooling Layer (readded)	-0.2%	-0.6%	-0.2%	-0.3%	36	0.02	5243	0.23
4x4 Pooling → 2x2 Pooling	-2.2%	0.4%	0.0%	-1.2%	73	0.02	5243	0.37
Late Fusion (removed)	-13.5%	-11.5%	-13.6%	-14.5%	78	0.01	2773	0.18
Late Fusion (removed) 2x Number of Filters	-14.2%	-12.3%	-16.6%	-16.5%	108	0.04	10343	0.19
Max Pooling → Avg Pooling (amplitude)	-1.6%	-1.1%	-1.6%	-1.7%	63	0.02	5243	0.23

TABLE V

CLASSIFICATION PERFORMANCE OF LENET-2AP WITH VARIOUS FUSION STRATEGIES

LeNet-2AP Fusion Method	Accuracy	Precision	Recall	F1 Score
Input Fusion	58.4%	56.2%	55.7%	52.4%
Early Fusion	60.9%	59.4%	60.2%	56.2%
Late Fusion	71.9%	67.7%	69.3%	66.9%
Decision Fusion	68.0%	63.9%	64.9%	62.7%

TABLE VI

CLASSIFICATION RESULTS OF LENET-2AP USING I–Q INPUT WITH INPUT FUSION AND LATE FUSION

LeNet-2AP I-Q Input	Accuracy	Precision	Recall	F1 Score
Input Fusion	61.9%	60.7%	59.6%	57.4%
Late Fusion	56.4%	58.2%	56.9%	53.0%

TABLE VII

CLASSIFICATION PERFORMANCE OF LENET-2AP WITH VARIOUS POOLING STRATEGIES

LeNet-2AP Pooling Method	Accuracy	Precision	Recall	F1 Score
Max Pooling (both branches)	66.9%	64.5%	65.1%	62.2%
Avg Pooling (amplitude) Max Pooling (phase)	62.6%	62.2%	61.7%	58.2%

TABLE VIII

CLASSIFICATION PERFORMANCE LENET-2AP WITH ISOLATED AMPLITUDE AND PHASE CHANNELS

Model	Accuracy	Precision	Recall	F1 Score
LeNet-2AP (amplitude)	63.2%	62.4%	62.0%	59.0%
LeNet-2AP (phase)	58.9%	59.4%	58.3%	54.6%

This highlights the value of the phase in resolving ambiguities that amplitude data alone cannot address.

C. Ablation Study

Table IV presents the impact of each ablated modification on classification performance, compared with the full LeNet-2AP model. Starting with the activation function, switching from ReLU to tanh decreases classification metrics by an average of 5%. The sparsity property of ReLU reduces the risk of overfitting, leading to more stable performance. Similarly, the removal of weight standardization results in a significant decrease in classification performance, with an average change of 7%. The weight standardization can be seen to greatly stabilize training, as its removal more than doubles the number of epochs required to reach the best performance. Consequently, this leads to a substantial increase to the total training time.

Reverting from the GAP layer to fully connected layers results in a classification performance reduction of approximately 10%. Dense connections, due to their complexity, significantly increase the risk of overfitting on small datasets. In contrast, GAP effectively summarizes the information from the convolutional layers into a more generalizable form. Another key advantage of the GAP layer is its ability to reduce model size and parameter count by three orders of magnitude. This occurs because dense layers require substantial memory to store the weights and biases of each neuron. In a model as shallow as LeNet-5, where the number of convolutional kernels is relatively small, dense layers dominate the parameter count. Replacing them with GAP in LeNet-2AP

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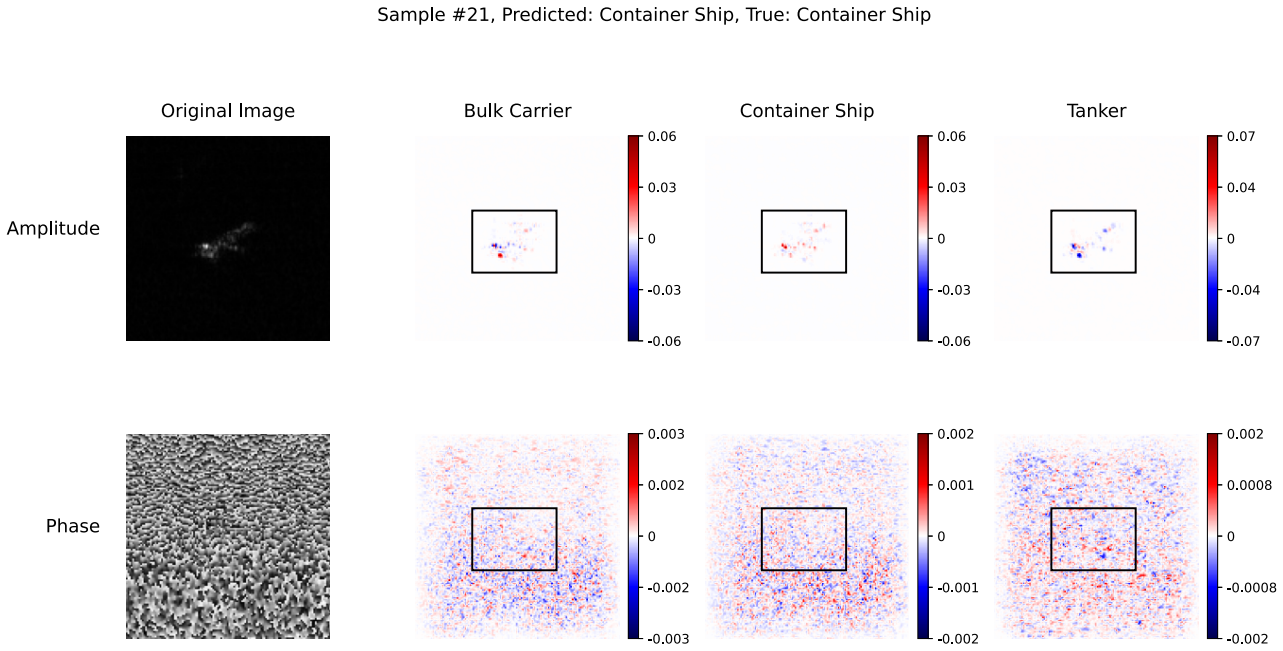


Fig. 5. SHAP values for a correctly classified container ship. The first column shows the original image, while the second, third, and fourth columns display SHAP values indicating the contribution of each feature toward classifying the sample as a bulk carrier, container ship, or tanker, respectively. The top row represents the amplitude channel, and the bottom row represents the phase channel. Positive contributions are highlighted in red, while negative contributions are shown in blue.

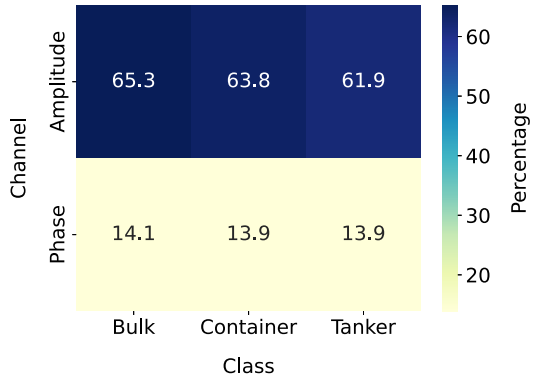


Fig. 6. Matrix highlighting the MCR, as defined in (11). It represents the percentage of the absolute SHAP value sum concentrated within the MER. Ship classes are displayed across columns and channels across rows.

not only mitigates overfitting but also significantly enhances efficiency.

Reintroducing the second pooling layer from the original LeNet-5 model has little impact on performance, resulting in an average decrease of 0.3% in classification metrics. This is likely because the feature maps after the second convolutional layer are already spatially small at 27×27 pixels, making additional downsampling prior to the GAP layer largely redundant. Similarly, reverting from 4×4 pooling to the original 2×2 pooling slightly decreases accuracy and $F1$ score by 2.2% and 1.2%, respectively. The larger pooling kernel reduces the spatial dimensions more aggressively, allowing the GAP layer to operate on a smaller and more focused feature map, which can enhance the summarization of information. The most significant advantage of using a larger pooling area is the improvement in computational efficiency, as the inference time per sample decreases from 0.37 to 0.25 ms.

The removal of the late fusion mechanism results in the most significant decrease among the ablated components, reducing classification performance by approximately 14%. Here, the amplitude and phase channels are concatenated at the input level and processed in a single branch. Although this modification shortens the inference time and halves the number of parameters, the model performs even worse than the baseline LeNet-5. To confirm that the observed decrease in performance is due to the absence of late fusion itself and not merely the reduction in filter count, a second ablation was performed with the same number of filters as LeNet-2AP (12 and 32 in the first and second convolutional layers, respectively). The classification results can be seen to decline within a similar margin, demonstrating that that the primary advantage of late fusion lies in its capacity to enable independent, modality-specific feature extraction before integration, rather than merely increasing the filter count.

To empirically validate the design choice of late fusion, a comparative analysis of alternative fusion strategies is summarized in Table V. Alongside late fusion and input fusion (defined in the previous paragraph), two additional fusion strategies were assessed.

1) *Early Fusion:* The two branches are fused after the first convolutional, pooling, and dropout layers.
2) *Decision Fusion:* Independent predictions from amplitude and phase branches are combined at the decision level via softmax score averaging.

The analysis reveals that late fusion achieves the highest performance, reaffirming the advantage of independently specialized branches for each modality. Decision fusion, which

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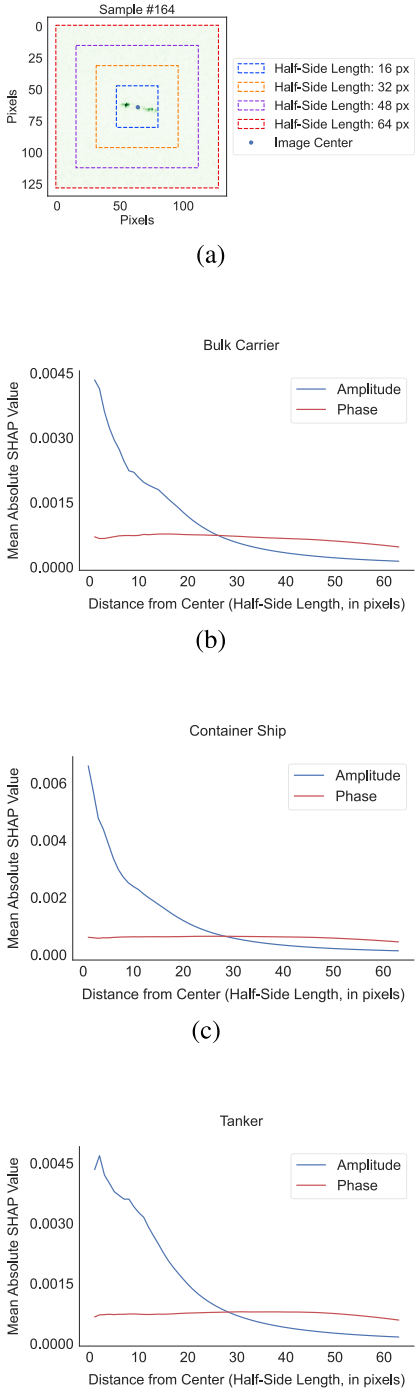


Fig. 7. (a) Example image overlaid with concentric squares centered on the ship, illustrating the regions used in the subsequent subplots. (b)–(d) DCF, as defined in (12), for bulk carriers, container ships, and tankers, respectively. Each plot shows the mean of the absolute SHAP values for the amplitude (blue) and phase (red) channels, averaged over all samples in the corresponding class, as a function of distance from the image center.

retains the independent processing capability, outperforms input and early fusion. Nonetheless, the absence of a learnable joint representation limits its overall effectiveness. Early fusion achieves better performance than input fusion, likely because it allows the model to partially process amplitude and phase features independently before combining them, offering a degree of feature extraction specialization prior to fusion. Collectively, this indicates that late fusion provides the optimal balance by allowing maximum independent feature specialization while retaining the capacity for a learnable, integrated representation prior to classification.

Further insight is provided in Table VI, which presents the classification results of LeNet-2AP with input fusion and late fusion using the I–Q input. Interestingly, in contrast to the amplitude–phase input, the input fusion yields higher classification metrics than late fusion. This can be attributed to the similarity between I and Q images in their representation of information, analogous to RGB channels in optical images. Concatenating the channels before the first convolutional layer allows the network to leverage both sources in a complementary manner. This outcome highlights a key principle: while analogous channels like I–Q benefit from being fused early, distinct modalities like amplitude and phase require separate, specialized processing to be effective. The degraded performance of LeNet-2AP on the I–Q input is a direct consequence of this principle. The model was specifically designed to process amplitude and phase as distinct modalities, a context where late fusion is ideal. This specialized design, however, becomes a liability when applied to the analogous I–Q input, leading to a lower performance.

Finally, the impact of applying max pooling to the amplitude branch is assessed by reverting it to average pooling. This modification results in a classification performance decrease of approximately 1.5%. The result indicates that the amplitude branch benefits from max pooling, as it prioritizes the propagation of features belonging to prominent scatterers. To further validate that the phase branch benefits specifically from average pooling, Table VII presents results for two additional configurations: applying max pooling to both branches and applying max pooling exclusively to the phase branch while retaining average pooling for the amplitude branch. In the former case, the $F1$ score decreases by nearly 5%, while in the latter, it drops by roughly 9%. This suggests that the cyclical and unwrapped nature of the phase is significantly disrupted when only the maximum feature is propagated. Instead, it derives its usefulness from the cumulative effect of neighboring values, which is effectively captured through average pooling.

The performance of each branch of LeNet-2AP in isolation is evaluated and summarized in Table VIII. The amplitude branch shows a decrease of around 9% in accuracy and 8% $F1$ score compared with the dual-channel LeNet-2AP model. The phase branch performs worse by approximately 3%–4% across all classification metrics compared with the amplitude. These results indicate that the discriminative information present in the phase is not only complementary to the amplitude but also capable of supporting classification independently, achieving a comparable level of performance. In addition, it demonstrates that the integration of both sources through a model specifically adapted to their distinct characteristics significantly enhances the overall classification performance.

To understand the results at the class level, confusion matrices are presented in Fig. 4 for LeNet-2AP using the combined amplitude–phase input, as well as each individually.

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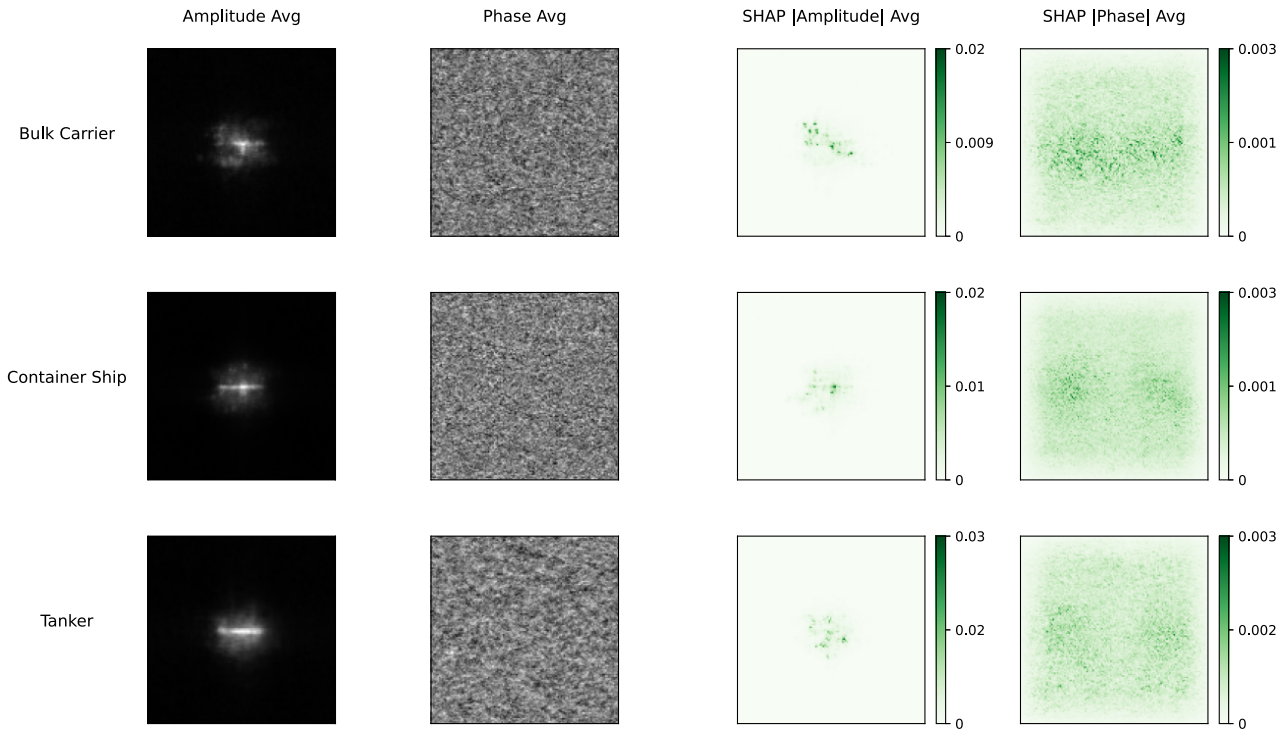


Fig. 8. First two columns display the pixelwise average amplitude and phase images across all samples for bulk carriers (first row), container ships (second row), and tankers (third row). The third and fourth columns display the CI, as defined in (13), representing the mean of the absolute SHAP values for the amplitude and phase channels, respectively, averaged over all predictions for that class.

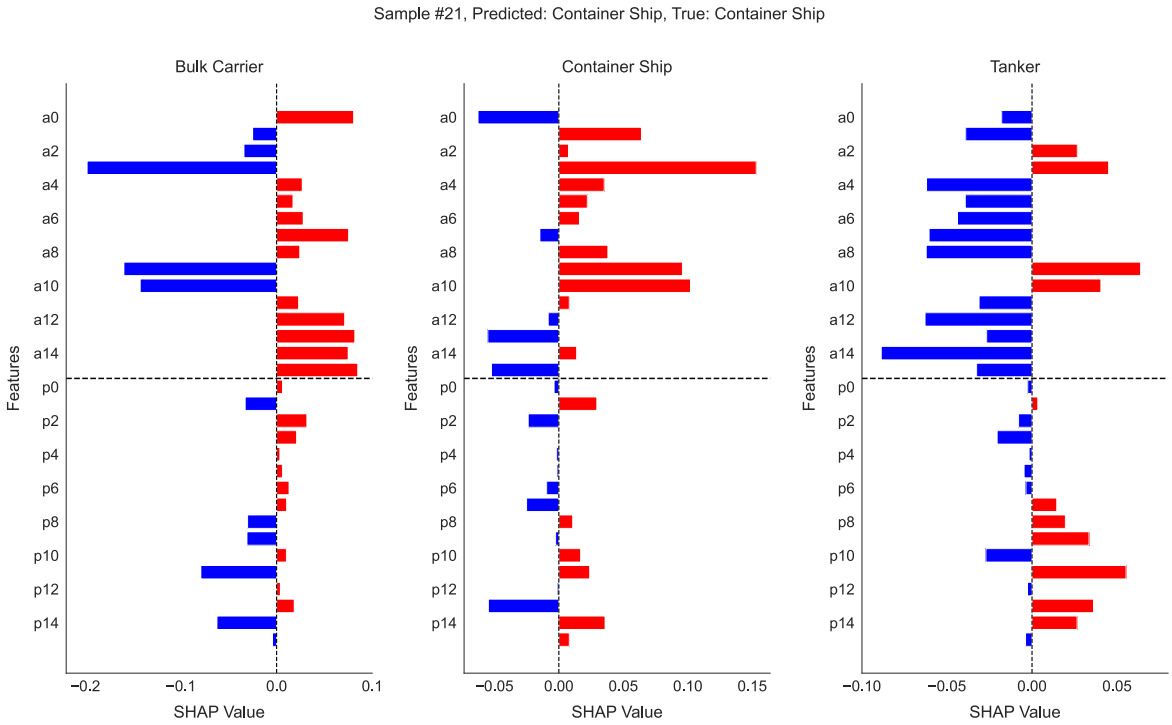


Fig. 9. SHAP values for the same sample shown in Fig. 5, calculated using the concatenated GAP vector as input instead of the raw image. The three subplots, from left to right, represent the feature contributions toward predicting the sample as a bulk carrier, container ship, and tanker, respectively. Elements a0–a15 correspond to GAP features from the amplitude channel, while p0–p15 represent those from the phase channel. Positive contributions are shown in red, and negative contributions are shown in blue. The dashed horizontal line marks the separation between amplitude and phase features, while the vertical line indicates an SHAP value of 0.

The dual-channel model performs best on tankers, achieving an accuracy of approximately 78%, and worst on bulk carriers, with an accuracy of 59%. Misclassifications in the bulk carrier class appear to stem equally from the other two classes, while for container ships, there is a predominance of false positives predicted to be bulk carriers. This pattern suggests that the

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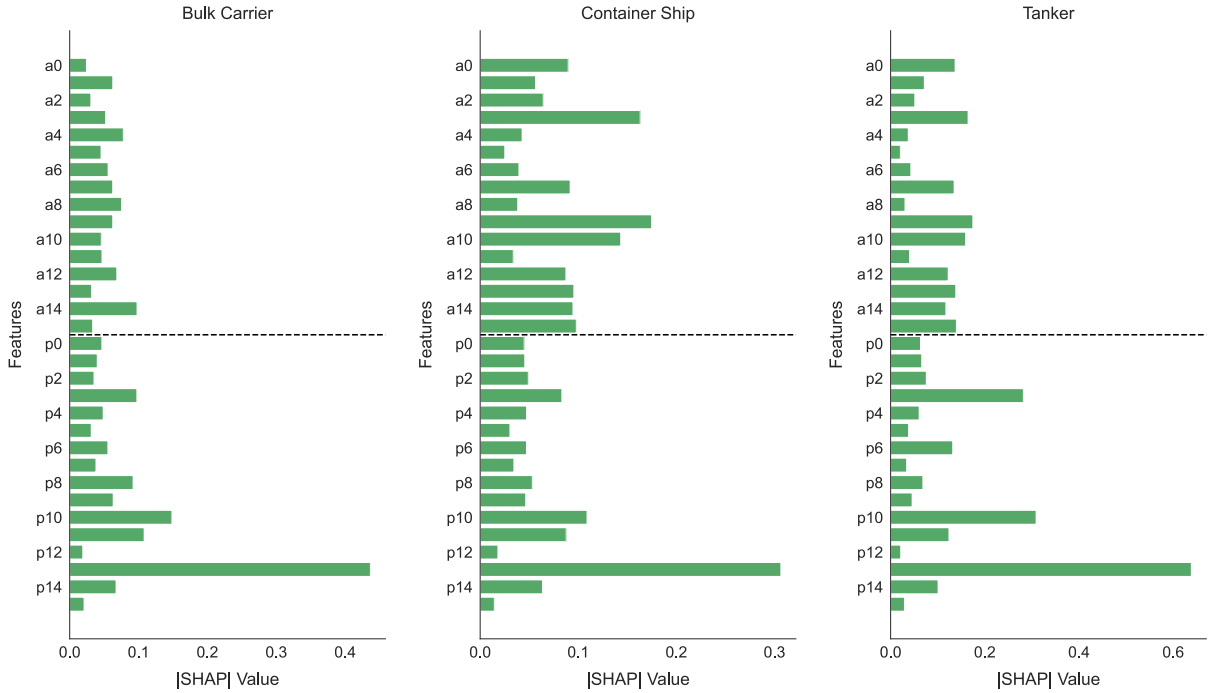


Fig. 10. CG, as defined in (15), representing the mean of the absolute SHAP values for the amplitude and phase GAP features, averaged over all predictions for samples in (left) bulk carrier, (middle) container ship, and (right) tanker classes. Feature labels and horizontal dashed line are as described in Fig. 9.

model struggles to distinguish features specific to bulk carriers, likely due to overlapping scattering characteristics with the other classes.

For the amplitude-only model, a decrease in accuracy is observed across all classes. The bulk carrier class is least affected, with a decrease of only 2%, whereas container ships and tankers experience reductions of 8% and 11%, respectively. In the phase-exclusive model, the accuracies also decline relative to the dual-channel model. However, the container ship class is impacted the most, with accuracy dropping by 14%, while the tanker class experiences only a minor decrease of 2%. These findings suggest that the phase plays a critical role in the classification of tankers, while the amplitude appears to be the dominant source of information for the bulk carrier class. This difference may arise from the unique superstructures of each ship type, which influence how SAR signals interact with their surfaces and contribute to amplitude and phase information. In bulk carriers, the dominant sources of backscatter are the numerous dihedral and trihedral structures supporting the hatch covers [58]. Nonetheless, their influence on the phase remains less clear. One possible explanation is that the phase modulations introduced by these structures are highly complex and inconsistent, making them difficult for the CNN to generalize from a limited dataset. Alternatively, it could be that the phase effects from these structures are inherently less pronounced or less informative for classification. Tankers, on the other hand, feature a dense network of pipelines arranged across their decks. These structures appear to contribute significantly to phase-based classification accuracy. One hypothesis is that the dense scattering mechanisms of these pipelines create intricate but informative phase modulations, which the model effectively leverages despite the small number of samples.

A possible explanation for why container ships benefit the most from the synergy of amplitude and phase channels lies in the variability of container arrangements on their decks. When containers are fully and evenly stacked, they create a flat, uniform surface, which limits cavity-like structures that would otherwise trap radar waves and induce significant phase modulation. In contrast, irregular or partially stacked arrangements introduce more varied geometries, potentially creating cavity-like scattering mechanisms that enhance phase information. This variability suggests that neither amplitude nor phase consistently dominates across different container arrangements, and the combination of both channels allows the model to adapt to these structural variations more effectively. Overall, LeNet-2AP demonstrates its ability to maximize the complementary strengths of amplitude and phase information, achieving superior performance across all classes compared with using either channel or in isolation.

D. Interpretability Study

To provide an intuitive understanding of the results obtained through SHAP analysis, a random sample is depicted in Fig. 5. This example, representing a correctly identified container ship, includes the original images for both the amplitude and phase channels alongside the SHAP values for each predicted class. In the amplitude channel, there is a notable concentration of positive SHAP values for the container ship prediction, while the other two classes are predominantly composed of negative values. In addition, the pixels with the largest SHAP magnitudes closely correspond to the ship's signature in the amplitude image. In contrast, the phase image presents a more complex pattern, where these contributions are distributed across the entire image rather than being localized to the ship's

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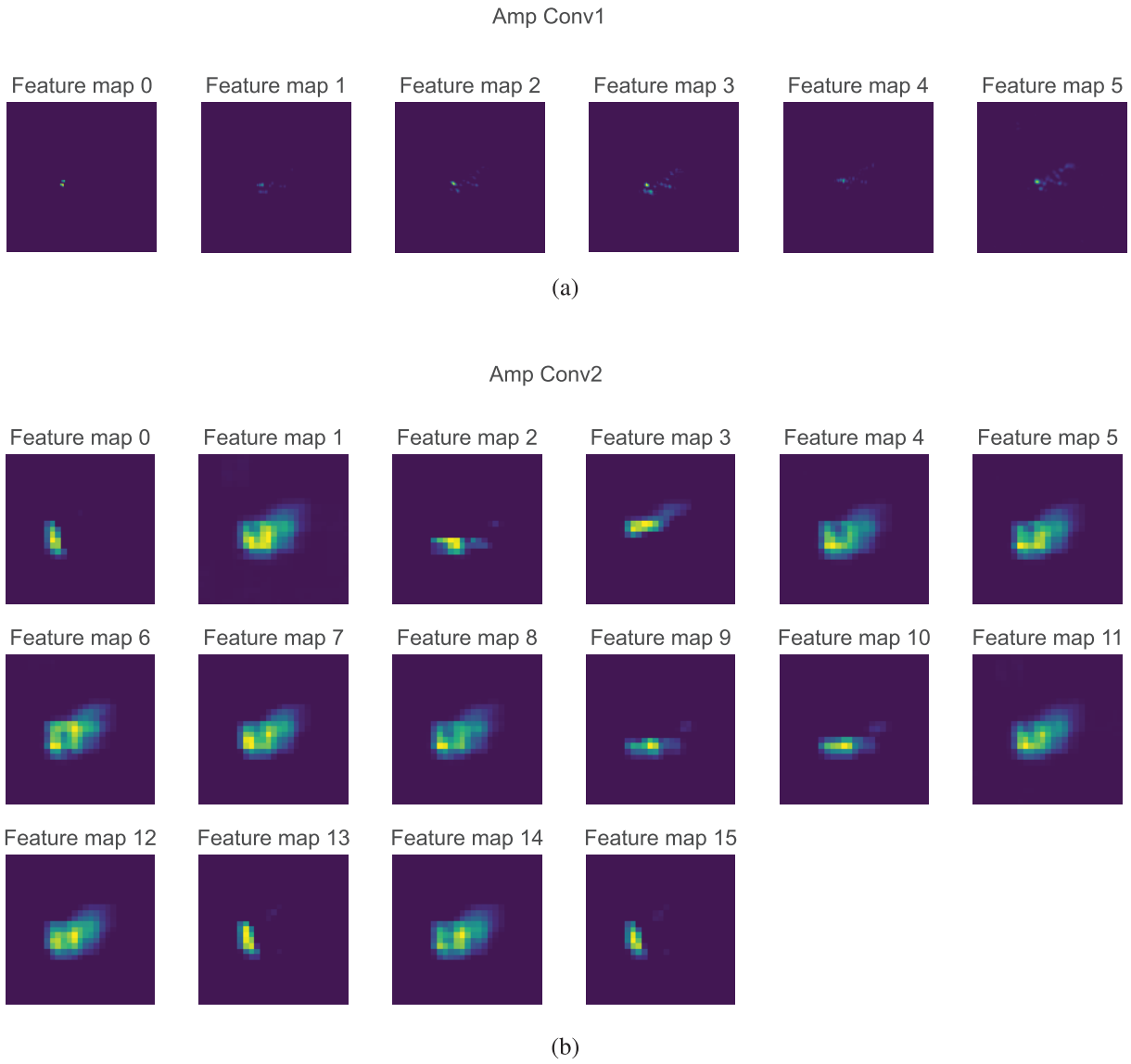


Fig. 11. Feature maps derived from the amplitude branch of the same sample as shown in Figs. 5 and 9. (a) Features extracted from the first convolutional layer. (b) Features extracted from the second convolutional layer.

MER. For the bulk carrier class, the bottom half of the phase image shows a dominance of negative contributions with an average SHAP value of -3×10^{-5}, whereas the top half exhibits a positive influence, averaging 1×10^{-5}. In contrast, for the tanker class, the bottom half shows a pronounced positive contribution with an average SHAP value of 2×10^{-5}, while the top half displays a slightly negative contribution of -1×10^{-5}. It is also noted that the phase SHAP values are an order of magnitude less than those in the amplitude channel, despite it having a greater number of high-contributing pixels.

To validate the generalization of the observed results, the MCR values presented in Fig. 6 are examined, providing a quantitative summary of the model's focus across different classes. Minor variations are evident in the amplitude channel, with interclass differences of approximately 3%. In contrast, the phase channel demonstrates minimal interclass variation, indicating a high degree of consistency. The most notable observation is the significant disparity between the two channels, with a gap of roughly 50 percentage points. On average, around two-third of the high-contributing pixels in the amplitude channel are located within the MER. This is consistent with the expectation that the amplitude predominantly captures the ship's structural features, which are spatially distinct from the background. Conversely, only 14% of the high-contributing pixels in the phase channel fall within the MER. This reinforces the observations from Fig. 5 that phase information does not strictly align with the spatial region of the amplitude signature.

Building on these findings, Fig. 7 presents line plots of the DCF as a function of distance from the center of the ship. A consistent trend is observed across all three classes, where amplitude contributions exhibit a sharp initial decrease near the center before progressively approaching 0, resembling an exponential decay. This behavior aligns with the earlier observation that the amplitude channel focuses on the ship's structural features within the MER. The steep initial

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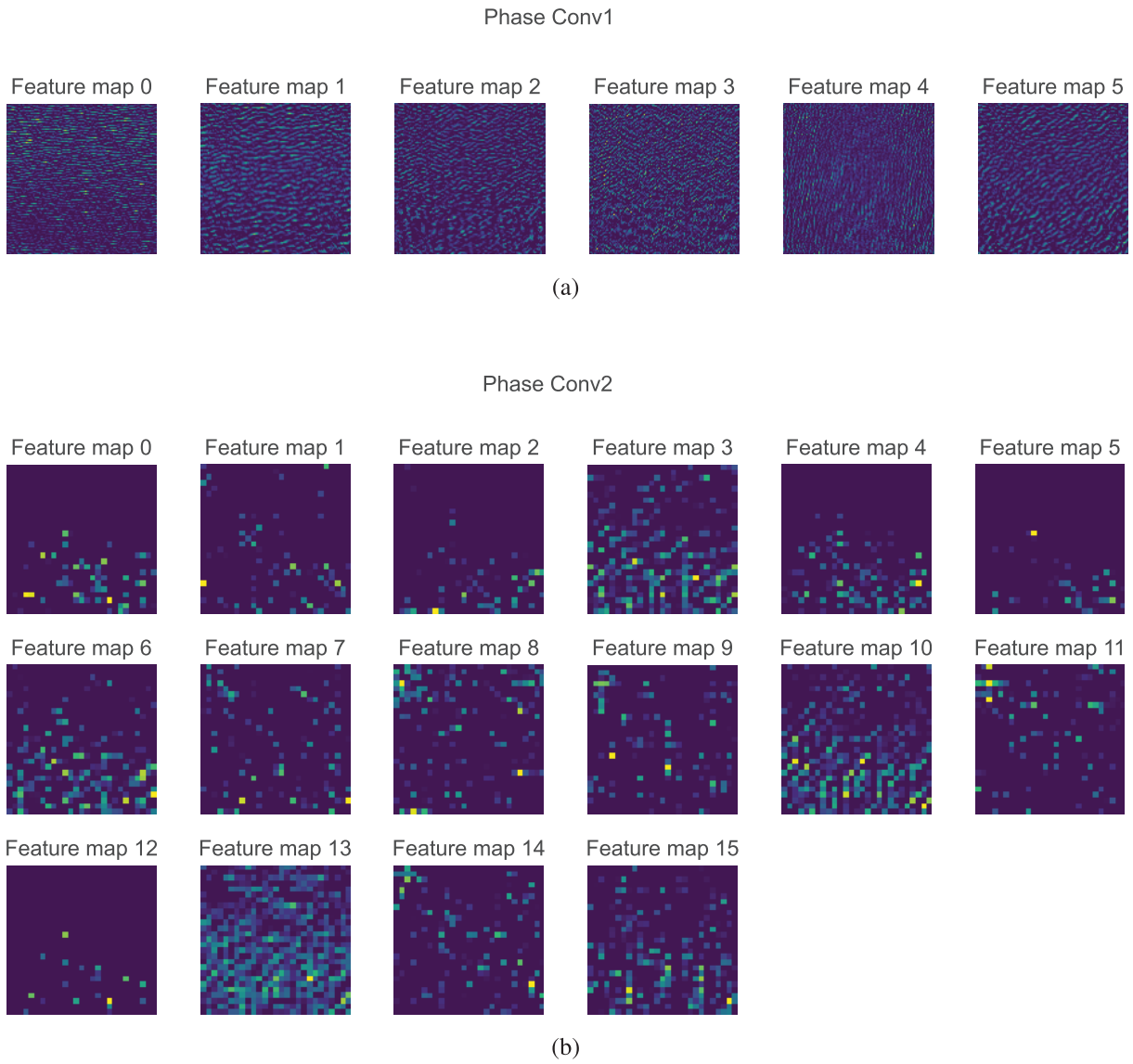


Fig. 12. Feature maps derived from the phase branch of the same sample as shown in Figs. 5 and 9. (a) Features extracted from the first convolutional layer. (b) Features extracted from the second convolutional layer.

decline corresponds to the inclusion of background pixels, which contribute minimally to the classification, reducing the mean contribution as the squares expand. Beyond 40 pixels from the center, the contributions stabilize as background pixels predominate. In contrast, phase contributions remain relatively uniform across the analyzed region, with a slight downward trend emerging around the 50-pixel mark for all classes. These results suggest that the phase, while more spatially dispersed than the amplitude, eventually transitions to being dominated by background influences, with contributions diminishing beyond a certain distance from the ship's center. This mirrors the behavior of the amplitude channel but occurs more gradually over a larger spatial range.

To visually summarize the results, averaged images of amplitude and phase are presented for each class in Fig. 8, alongside the CIs. First, while the absolute SHAP values in the phase channel are an order of magnitude lower than those in the amplitude channel, a significantly larger number of pixels exhibit relatively high SHAP values in the phase channel. In contrast, the amplitude channel contains only a small subset of such pixels, consistent with the example in Fig. 5. Second, the amplitude SHAP values are predominantly concentrated within the MER, highlighting the strong spatial localization of amplitude contributions. In contrast, phase SHAP values are more broadly distributed across the image, reinforcing the observation from the MCR analysis that phase contributions extend beyond the MER. Third, an apparent decrease in phase SHAP values toward the image borders aligns with the results from the DCF analysis in Fig. 7, suggesting a spatial limit beyond which background influences dominate the phase contributions. Finally, the SHAP composites reveal a horizontal pattern in the phase contributions, where high-valued pixels are distributed symmetrically on either side of the center. This observation could be attributed to the asymmetric spatial resolution of Sentinel-1 SLC products (3 m in range and 22 m in azimuth), suggesting that phase contributions are influenced by resolution. Such an interpretation aligns with theory and prior literature [12], which indicate that higher resolution

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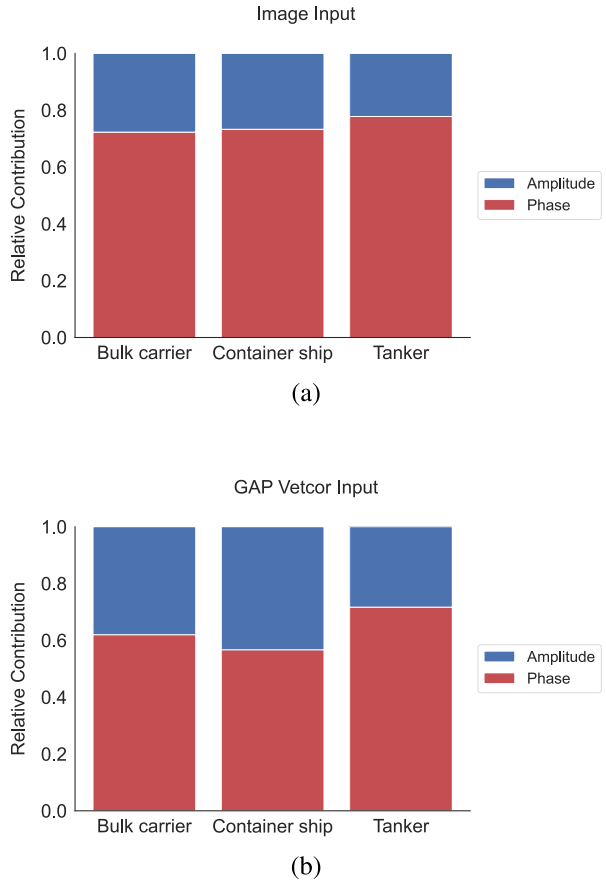


Fig. 13. (a) RC and (b) RCG, as defined in (14) and (16), respectively. For each ship class (bulk carrier, container ship, and tanker), red bars represent the contribution of the amplitude channel, and blue bars represent the contribution of the phase channel.

amplifies deviations from the fully developed speckle model, thereby making phase effects more pronounced.

An alternative perspective is provided by the GAP SHAP analysis, which shifts the focus from individual pixels to feature-level contributions. Fig. 9 presents the SHAP analysis for the same sample shown in Fig. 5, but focusing exclusively on the final GAP and softmax layers. In this context, the features correspond to the 32 elements of the GAP vector. The highest positive contributions are attributed to amplitude features of the correctly identified container ship class, while the corresponding features for the other two classes exhibit strong negative contributions. The phase-derived features appear generally smaller in magnitude, suggesting that, for this sample, amplitude features contribute more significantly to the classification outcome than phase features. However, when examining Fig. 10, which presents the CG per class, phase features display comparable or even higher absolute SHAP values than amplitude features. Notably, feature p13 consistently exhibits the highest magnitude across all three classes, suggesting its importance in the classification process. This divergence between sample-specific and class-level feature contributions demonstrates that while individual predictions may disproportionately rely on one modality, broader trends derived from aggregated metrics provide a more reliable characterization of class-wide model behavior. This reinforces the rationale for incorporating aggregate-level explainability analyses, as relying solely on isolated examples could lead to misleading or unrepresentative conclusions about model decision-making.

To better understand the feature-level patterns associated with p13, as well as the remaining features, the activation maps of the first and second convolutional layers from the amplitude and phase channels are displayed in Figs. 11 and 12, respectively. These maps are derived from the same sample shown in Figs. 5 and 9. For the amplitude channel, the strongest activations appear to be concentrated in the stern of the ship, demonstrating consistency across all 16 feature maps of the second convolutional layer. In contrast, the first-layer activations in the phase channel seem to capture directional patterns. For example, feature maps 0 and 1 emphasize horizontal patterns, while feature maps 4 and 5 highlight vertical and diagonal patterns, respectively. The second-layer activations in the phase channel appear more abstract, with patterns that are less localized and not confined to specific regions of the ship. Notably, feature map 13, from which p13 is derived, exhibits the most extensive activation distribution, spanning a broader area compared with other feature maps.

Finally, Fig. 13 illustrates the RC and RCG of the amplitude and phase channels for each class, based on the two SHAP approaches. The RC plot indicates a higher contribution of the phase channel across all three classes, consistently exceeding 70%. This can be attributed to the phase channel's larger number of contributing pixels compared with the amplitude channel. The RCG shows a similar trend, with the phase remaining the major contributor, albeit less significantly than in the full-model analysis. Furthermore, interclass variation is more pronounced in the GAP analysis, with phase contributions of 62%, 57%, and 72% for the bulk carrier, container ship, and tanker classes, respectively. These values align with the confusion matrices in Fig. 4, which show that the phase-exclusive model performs best at classifying tankers, reflecting the highest relative phase contribution for this class. In contrast, although the amplitude-exclusive model outperforms the phase-exclusive model for the other two classes, the dual-channel model consistently attributes a higher RC to the phase channel than to the amplitude channel. This indicates that the phase, despite being less effective in isolation, contributes significantly to the LeNet-2AP model across all classes.

V. Conclusion

This article has demonstrated that with a specialized architecture, the phase is a powerful and nonredundant resource for single-polarization SAR ship classification. This was achieved through the introduction of LeNet-2AP as an efficient and lightweight architecture designed to effectively leverage the phase. The model achieved state-of-the-art performance with limited training data and demonstrated its effectiveness against established CNN models. Furthermore, a detailed ablation study highlighted the contribution of each modified component to the overall classification performance. The late fusion mechanism yielded the most significant improvement in classification accuracy, proving the value of processing the channels independently, while replacing the fully connected layers with GAP drastically reduced the model size and parameter count without compromising performance. Analysis of each channel in isolation revealed that the phase channel is particularly effective at identifying tankers.

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In addition, this article elucidated the role of the phase through the SHAP analysis. It was shown that the amplitude is strongly localized within MER, while the phase information are much more spatially dispersed. The DCF analysis further supported these findings, emphasizing that the two channels capture fundamentally different spatial characteristics. Although phase contributions extended further, the results indicated a spatial limit of 50 pixels beyond which background influences became predominant. Moreover, the CI analysis suggest a dependence of phase contributions on spatial resolution, with more pronounced effects observed in the range direction, corresponding to the higher spatial resolution in Sentinel-1 SLC acquisitions. Finally, a faster and more efficient alternative for evaluating the overall contribution of the two sources was presented, using the final two layers of LeNet-2AP instead of the full model. This approach yielded comparable results to the full model in terms of the overall contribution of amplitude and phase.

While this study novelly presented the phase as a source of comparable discriminative information to amplitude, several questions remain that warrant further investigation. The confusion matrices highlighted that tankers benefit the most from the synergy of the two sources, while bulk carriers showed minimal improvement compared with the amplitude-exclusive model. Future research will explore the influence of varying superstructures within each class and how their scattering mechanisms contribute to distinct phase patterns. In addition, it is critical to determine the spatial resolution at which phase contributions become significant and to confirm whether their significance increases at higher resolutions. Moreover, understanding the relationship between target size and phase effects is essential, including identifying the minimum target size necessary to observe meaningful phase contributions.

Further work should also investigate the robustness of these findings under challenging operational conditions. For example, determining if rough sea conditions diminish phase effects similar to amplitude, or if phase contributions demonstrate independence, potentially making phase a more reliable source in such scenarios. Quantifying the precise extent to which this spatial influence persists is also crucial, particularly in scenarios involving dense shipping lanes or ships docked at port.

The implications of this work, however, extend beyond maritime applications. Addressing all the aforementioned questions will provide a deeper understanding of the role of phase, but the key phenomenon underpinning its value should, in theory, extend to other man-made extended targets such as oil rigs, wind farms, or even urban structures. In such settings, understanding how phase features from one target interact with nearby objects becomes imperative. This highlights the need for a better alternative to account for limitations in the fully developed speckle model, especially when its core assumptions are violated by dominant scatterers. Overall, this study not only demonstrates the value of phase information for ship classification but also raises important questions about its potential role in broader classification tasks in single-polarization SAR.

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